Gravitational field strength

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a gravitational field to predict the force on a test mass.

SUCCESS CRITERIA - I can...

1. Represent field direction with arrows.
2. Use g=F/m and F=mg consistently.
3. Distinguish source field from the mass placed in it.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

An object weighs 29.4 N where g=9.8 N/kg. Find its mass and its weight in a field of 5 N/kg.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $g = F_g/m = GM/r^2$.
Outside a spherical source of mass M; r is distance from its center. Field direction is toward the source.

Worked reasoning: A 2 kg test object feels 6 N toward a source. The local field is g=6/2=3 N/kg. A 4 kg test object at that point feels 12 N.

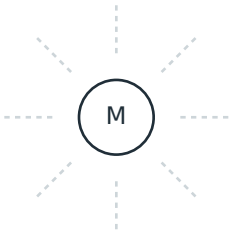
Your turn: Draw the field arrow and the two possible force arrows. Which quantity belongs to the location rather than the test mass?

Test mass (kg)	Force (N)	g (N/kg)
2	6	___
4	12	___

Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw field arrows at three locations around the source. Which way would a small test mass accelerate?



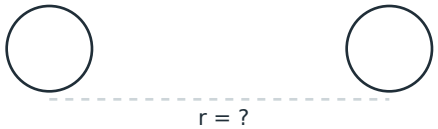
Add arrowheads to show field direction.

The center marks the source; add arrowheads on the surrounding lines to show field direction.

Marzano check - circle: ① ② ③

04 Apply the physics

A 0.20 kg test mass experiences a 0.60 N gravitational force. Find the field strength. The source has GM = 12 m^3/s^2; find the test mass's center-to-center distance from the source.



Source: GM = 12 Test mass: 0.20 kg

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A student says a heavier test mass falls faster because it feels more gravitational force. Use F=mg and a=F/m to respond.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

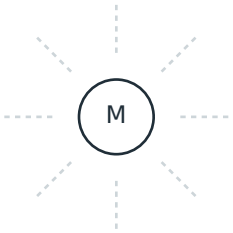
A 4 kg block slides down a 30° ramp with 11.6 N friction up the ramp. Use g=9.8 m/s². Find along-ramp acceleration and normal force.
Draw the forces on the chosen object and label their sources.

Add and label force arrows.

Marzano check - circle: ① ② ③

07 Exit check

Outside a spherical source, field is 8 N/kg at r. Find field at 2 r and force on 0.50 kg there.



Add arrowheads to show field direction.

Marzano check - circle: ① ② ③

08 STRETCH: test the model

An exterior field is 8 N/kg at distance r from a spherical source. Where is it 2 N/kg? What force acts there on a 0.25 kg mass?

Location	Test mass (kg)	Field (N/kg)	Force (N)
1	____	____	____
2	____	____	____

Marzano check - circle: ① ② ③